

- Solutions -

AP Physics Ch 11 and 12 – Rotational Motion Multiple Choice Problem Set

Rotational Motion and Angular Displacement Angular Velocity and Angular Acceleration

- B 1. What is the angular speed in rad/s of the second hand of a watch?
- (a) 1.7×10^{-3} rad/s (c) 0.02 rad/s (e) 60 rad/s
(b) 0.10 rad/s (d) 6.28 rad/s

$$\frac{2\pi \text{ rad}}{60 \text{ sec}} = 0.105 \text{ rad/s}$$

- C 2. A wheel with a 0.10-m radius is rotating at 35 rev/s. It then slows uniformly to 15 rev/s over a 3.0-s interval. What is the angular acceleration of a point on the wheel?
- (a) -2.0 rev/s^2 (c) -6.7 rev/s^2 (e) -17 rev/s^2
(b) 0.67 rev/s^2 (d) 42 rev/s^2

$$\omega_f = \omega_i + \alpha t$$

$$15 = 35 + \alpha(3) \quad \alpha = -6.7 \frac{\text{rev}}{\text{s}^2}$$

The Equations of Rotational Kinematics

- C 3. During the spin-dry cycle of a washing machine, the motor slows from 95 rad/s to 30 rad/s while the turning the drum through an angle of 402 radians. What is the magnitude of the angular acceleration of the motor?
- (a) 64 rad/s^2 (c) 10 rad/s^2 (e) 1.0 rad/s^2
(b) 32 rad/s^2 (d) 20 rad/s^2

$$\omega_f^2 = \omega_i^2 + 2\alpha\Delta\theta$$

$$30^2 = 95^2 + 2\alpha(402) \quad \alpha = -10.1 \text{ rad/s}^2$$

- A 4. An airplane engine starts from rest; and 2 seconds later, it is rotating with an angular speed of $t = 2 \text{ sec} \cdot \left(\frac{1 \text{ min}}{60 \text{ s}}\right) = 0.033$ 300 rev/min. If the angular acceleration is constant, how many revolutions does the propeller undergo during this time?

- (a) 5 (c) 50 (e) 600
(b) 10 (d) 300

$$\theta_f = \theta_i + \omega_i t + \frac{1}{2} \alpha t^2 = 5$$

$$\omega_f = \omega_i + \alpha t$$

$$300 = 0 + \alpha(0.033)$$

$$\alpha = 9090$$

Questions 5 through 7 pertain to the statement below:

A grindstone of radius 4.0 m is initially spinning with an angular speed of 8.0 rad/s. The angular speed is then increased to 10 rad/s over the next 4.0 seconds. Assume that the angular acceleration is constant.

- D 5. What is the average angular speed of the grindstone?

- (a) 0.5 rad/s (c) 4.5 rad/s (e) 18 rad/s
(b) 2.0 rad/s (d) 9.0 rad/s

$$\omega_{\text{ave}} = \frac{10 + 8}{2} = 9 \frac{\text{rad}}{\text{s}} \quad \text{or} \quad \omega = \frac{\Delta\theta}{\Delta t} = \frac{36}{4} = 9$$

- A 6. What is the magnitude of the angular acceleration of the grindstone?

- (a) 0.50 rad/s^2 (c) 4.5 rad/s^2 (e) 18 rad/s^2
(b) 2.0 rad/s^2 (d) 9.0 rad/s^2

$$\omega_f = \omega_i + \alpha t \quad 10 = 8 + \alpha(4) \quad \alpha = 0.5$$

- D 7. Through how many revolutions does the grindstone turn during the 4.0-second interval?

- (a) 0.64 (c) 4.0 (e) 36
(b) 3.8 (d) 5.7

$$\Delta\theta = \omega_i t + \frac{1}{2} \alpha t^2 = 8(4) + \frac{1}{2}(0.5)(4)^2 = 36 \text{ rad} \cdot \frac{1 \text{ rev}}{2\pi \text{ rad}} = 5.7$$

A

8. A spinning disc rotating at 130 rev/min slows and stops 31 s later. How many revolutions did the disc make during this time?

(a) 34
(b) 67

(c) 8.4
(d) 17

(e) 4.2

$$\omega_f = \omega_i + \alpha t$$

$$0 = 130 + \alpha(.517) \quad \alpha = -251.45$$

$$\Delta\theta = \omega_i t - \frac{1}{2} \alpha t^2$$

$$= 130(.517) - \frac{1}{2} (251.45)(.517)^2$$

$$= 33.61$$

Angular Variables and Tangential Variables

D

9. What is the tangential speed of a lug nut on a wheel of a car if the lug nut is located 0.114 m from the axis of rotation; and the wheel is rotating at 6.53 rev/sec?

(a) 0.745 m/s
(b) 1.49 m/s

(c) 2.98 m/s
(d) 4.68 m/s

(e) 9.36 m/s

$$6.53 \frac{\text{rev}}{\text{sec}} \cdot \frac{2\pi \text{ rad}}{1 \text{ rev}} = 41.03$$

$$v = \omega r = 41.03(.114) = 4.68 \text{ m/s}$$

E

10. What is the tangential speed of Nairobi, Kenya, a city near the equator? The earth makes one revolution every 23.93 h and has an equatorial radius of 6380 km.

(a) 74.0 m/s
(b) 116 m/s

(c) 148 m/s
(d) 232 m/s

(e) 465 m/s

$$23.93 \text{ hr} \cdot \frac{3600 \text{ sec}}{1 \text{ hr}} = 86148 \text{ sec}$$

$$v = \frac{2\pi r}{t} = \frac{2\pi (6.38 \times 10^6)}{86148} = 465 \text{ m/s}$$

D

11. On an amusement park ride, passengers are seated in a horizontal circle of radius 7.5 m. The seats begin from rest and are uniformly accelerated for 21 seconds to a maximum rotational speed of 1.4 rad/s. What is the instantaneous tangential speed of the passengers 15 s after the acceleration begins?

(a) 0.067 m/s
(b) 0.50 m/s

(c) 1.4 m/s
(d) 7.5 m/s

(e) 11 m/s

$$\omega_f = \omega_i + \alpha t \quad 1.4 = \alpha(21) \quad \alpha = .067$$

$$\omega_f = .067(15) = 1 \text{ rad/s}$$

$$v_{\text{tan}} = \omega r = 1(7.5) = 7.5 \text{ m/s}$$

Section 8.5 Centripetal Acceleration and Tangential Acceleration

C

12. A circular disk of radius 0.010 m rotates with a constant angular speed of 5.0 rev/s. What is the acceleration of a point on the edge of the disk?

(a) 0.31 m/s²
(b) 1.6 m/s²

(c) 9.9 m/s²
(d) 2500 m/s²

(e) zero m/s²

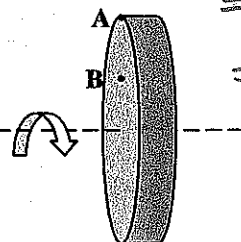
$$\omega = \text{const} \therefore \alpha = 0 \text{ and } a_{\text{tan}} = \alpha r = 0$$

$$a_{\text{rad}} = \frac{v^2}{r} = \omega^2 r = (5(2\pi))^2 (0.01) = 9.87$$

D

13. Two points are located on a rigid wheel that is rotating with decreasing angular velocity about a fixed axis. Point A is located on the rim of the wheel and point B is halfway between the rim and the axis. Which one of the following statements concerning this situation is true?

(a) Both points have the same centripetal acceleration. $a_{\text{rad}} = \omega^2 r$
(b) Both points have the same tangential acceleration. $a_{\text{tan}} = \alpha r$
(c) The angular velocity at point A is greater than that of point B. Same ω
(d) Both points have the same instantaneous angular velocity. Same ω
(e) Each second, point A turns through a greater angle than point B. Same angle



- B 14. The original Ferris wheel had a radius of 38 m and completed a full revolution (2π radians) every two minutes when operating at its maximum speed. If the wheel were uniformly slowed from its maximum speed to a stop in 35 seconds, what would be the magnitude of the tangential acceleration at the outer rim of the wheel during its deceleration?
- (a) 0.0015 m/s^2 (c) 0.54 m/s^2 (e) 6.8 m/s^2
 (b) 0.057 m/s^2 (d) 1.6 m/s^2

$$\omega_i = \frac{2\pi \text{ rad}}{120 \text{ sec}} = .0524$$

$$\omega_f = \omega_i + \alpha t$$

$$0 = .0524 + \alpha (35)$$

$$\alpha = .0015$$

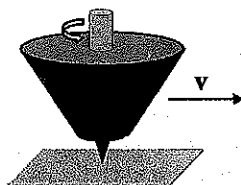
$$a = \alpha r$$

$$= .057 \frac{\text{m}}{\text{s}^2}$$

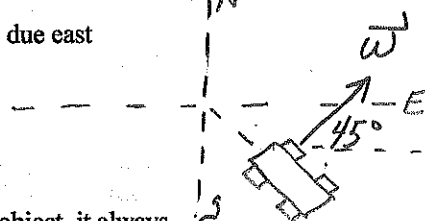
The Vector Nature of Angular Variables

- A 15. A top is spinning counterclockwise as shown in the figure. It is also moving to the right with a linear speed v . What is the direction of the angular velocity?

- (a) upward (d) right
 (b) downward (e) into the paper
 (c) left
- curl finger of right hand and stick thumb out*



- D 16. A car is moving along a horizontal road at a constant velocity that is directed 45° south of east. What is the direction of the angular velocity of the wheels of the car?
- (a) 45° south of west (c) 45° south of east (e) due east
 (b) 45° north of west (d) 45° north of east



The Effects of Forces and Torques on the Motion of Rigid Objects

- E 17. Complete the following statement: When a net torque is applied to a rigid object, it always produces a
- (a) constant acceleration. (d) constant angular momentum.
 (b) rotational equilibrium. (e) change in angular velocity.
 (c) constant angular velocity.

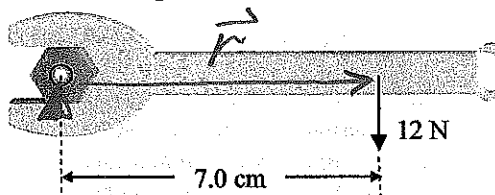
$$\vec{\tau}_{\text{net}} = I\alpha = I \frac{\Delta\omega}{\Delta t}$$

- B 18. A wrench is used to tighten a nut as shown in the figure. A 12-N force is applied 7.0 cm from the axis of rotation. What is the torque due to the applied force?

- (a) $0.58 \text{ N}\cdot\text{m}$
 (b) $0.84 \text{ N}\cdot\text{m}$
 (c) $1.71 \text{ N}\cdot\text{m}$
 (d) $14 \text{ N}\cdot\text{m}$
 (e) $58 \text{ N}\cdot\text{m}$

$$\vec{\tau} = \vec{r} \otimes \vec{F}$$

$$= r F_{\text{tan}} = .07 \text{ m} (12 \text{ N}) = .84 \text{ N}\cdot\text{m}$$



- A 19. A string is tied to a doorknob 0.79 m from the hinge as shown in the figure. At the instant shown, the force applied to the string is 5.0 N. What is the torque on the door?

- (a) $3.3 \text{ N}\cdot\text{m}$ (d) $0.84 \text{ N}\cdot\text{m}$
 (b) $2.2 \text{ N}\cdot\text{m}$ (e) $0.40 \text{ N}\cdot\text{m}$
 (c) $1.1 \text{ N}\cdot\text{m}$

$$\vec{\tau} = \vec{r} \otimes \vec{F}$$

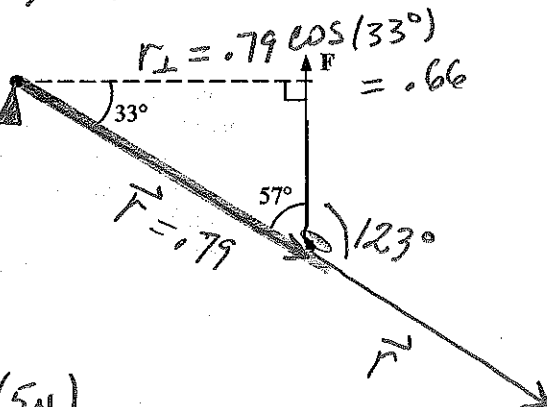
$$\tau = r_{\perp} F$$

$$= .79 (5) \sin 123^\circ$$

$$= 3.31 \text{ N}\cdot\text{m} (+\hat{z})$$

$$= (.66 \text{ m})(5 \text{ N})$$

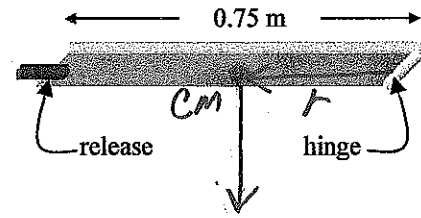
$$= 3.3 \text{ N}\cdot\text{m}$$



C

20. A uniform 13-kg trap door is oriented horizontally and hinged as shown. What is the magnitude of the torque on the door at the instant that the release is activated and the door can freely rotate?

- (a) 4.9 N · m (d) 72 N · m
(b) 9.8 N · m (e) 96 N · m
(c) 48 N · m



$$mg = 13(9.8) = 127 \text{ N}$$

$$\begin{aligned}\vec{\tau} &= \vec{r} \otimes \vec{F} \\ &= (0.375 \text{ m})(127 \text{ N}) \sin 90^\circ (+\hat{z}) \\ &= 47.6 \text{ N} \cdot \text{m}\end{aligned}$$

$$\sum \tau = 0$$

→ Rigid Objects in Equilibrium
Center of Gravity

D

21. Complete the following statement: A body is in translational equilibrium

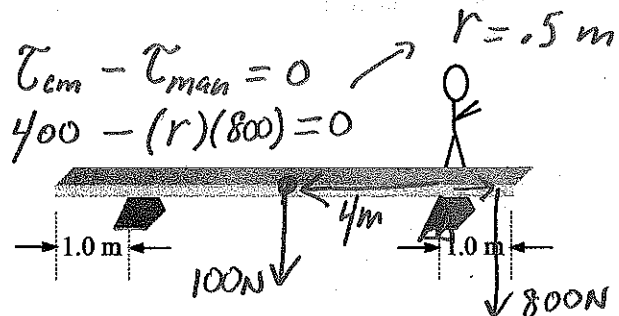
- (a) only if it is at rest.
(b) only if it is moving with constant velocity.
(c) only if it is moving with constant acceleration.
(d) if it is either at rest or moving with constant velocity.
(e) if it is moving with either constant velocity or constant acceleration.

$$F_{\text{net}} = ma = 0$$

C

22. A horizontal, 10-m plank weighs 100 N. It rests on two supports that are placed 1.0 m from each end as shown in the figure. How close to one end can an 800-N person stand without causing the plank to tip?

- (a) 0 m (d) 0.7 m
(b) 0.3 m (e) 0.9 m
(c) 0.5 m



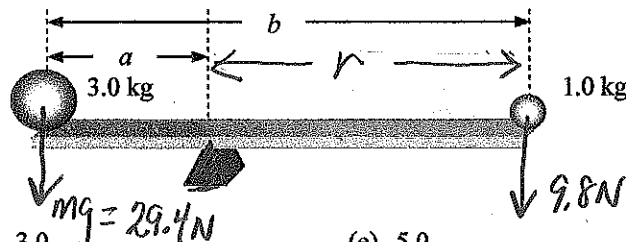
$$\begin{aligned}\tau_{\text{cm}} - \tau_{\text{man}} &= 0 \\ 400 - (r)(800) &= 0\end{aligned}$$

D

23. A 3.0-kg ball and a 1.0-kg ball are placed at opposite ends of a massless beam so that the system is in equilibrium as shown. Note: The drawing is not drawn to scale. What is the ratio of the lengths, b/a ?

- (a) 2.0
(b) 2.5

- (c) 3.0
(d) 4.0



- (e) 5.0

$$\tau_3 = \tau_1$$

$$\begin{aligned}a(29.4) &= r(9.8) \\ \frac{r}{a} &= 3\end{aligned}$$

$$\begin{aligned}r + a &= b \\ 3a + a &= b \\ 4a &= b \\ \frac{b}{a} &= 4\end{aligned}$$

B

24. One end of a rope is tied to the handle of a horizontally-oriented and uniform door. A force F is applied to the other end of the rope as shown in the drawing. The door has a weight of 145 N and is hinged on the right. What is the maximum magnitude of F for which the door will remain at rest?

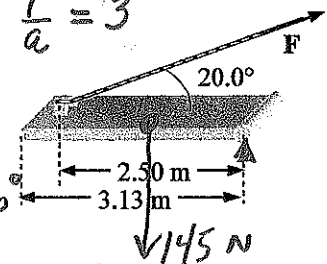
- (a) 145 N
(b) 265 N
(c) 381 N

$$\tau_{\text{cm}} = \left(\frac{3.13}{2}\right) 145$$

- (d) 424 N
(e) 530 N

$$\begin{aligned}\tau_F &= (2.5)(F) \sin 60^\circ \\ &= .855 F\end{aligned}$$

$$= 226.9 \text{ N} \cdot \text{m}$$



$$F = \frac{226.9}{.855} = 265.4 \text{ N}$$

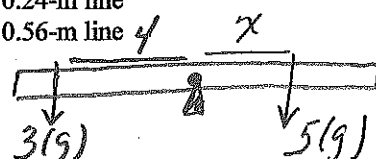
D

25. A meter stick is pivoted at the 0.50-m line. A 3.0-kg object is hung from the 0.10-m line. Where should a 5.0-kg object be hung to achieve equilibrium?

- (a) 0.06-m line
(b) 0.24-m line
(c) 0.56-m line

- (d) 0.74-m line

- (e) A 5.0-kg object cannot be used to result in equilibrium in this system.



$$\tau_3 = \tau_5$$

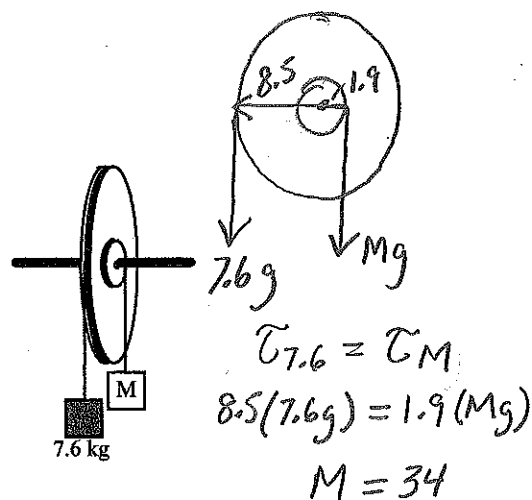
$$4(3g) = x(5g)$$

$$x = 2.4 \text{ cm}$$

A

26. In the drawing shown, the large wheel has a radius of 8.5 m. A rope is wrapped around the edge of the wheel and a 7.6 kg-box hangs from the rope. A smaller disk of radius 1.9 m is attached to the wheel. A rope is wrapped around the edge of the disk as shown. An axis of rotation passes through the center of the wheel-disk system. What is the value of the mass M that will prevent the wheel from rotating?

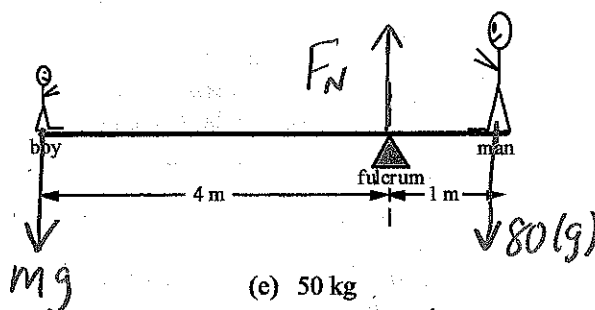
- (a) 34 kg (d) 3.8 kg
(b) 12 kg (e) 46 kg
(c) 1.7 kg



Questions 27 and 28 pertain to the situation described below:

An 80-kg man balances the boy on a teeter-totter as shown.

Note: Ignore the weight of the board.



27. What is the approximate mass of the boy?
(a) 10 kg (c) 40 kg
(b) 20 kg (d) 45 kg (e) 50 kg

28. What, approximately, is the magnitude of the downward force exerted on the fulcrum?

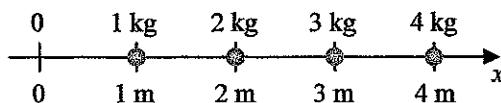
- (a) zero newtons (c) 600 N (e) 1000 N
(b) 100 N (d) 800 N

$$F_N = 20(9.8) + 80(9.8) = 980 \text{ N}$$

29. Which one of the following statements most accurately describes the center of gravity of an object?

- (a) It is the point where gravity acts on the object. — If object is point particle
(b) It is the point where all the mass is concentrated.
(c) It must be experimentally determined for all objects. — NO
(d) It is the point on the object where all the weight is concentrated.
(e) It is the point from which the torque produced by the weight of the object can be calculated. — Smallest possible I for an object

30. Consider four point masses located as shown in the sketch. The acceleration due to gravity is the same everywhere.



What is the x coordinate of the center of gravity for this system?

- (a) 2.0 m (c) 3.0 m (e) 3.8 m
(b) 2.7 m (d) 3.3 m

$$\bar{x}_{cm} = \frac{m_1x_1 + m_2x_2 + \dots}{M}$$

$$= \frac{1(1) + 2(2) + 3(3) + 4(4)}{(1+2+3+4)}$$

$$= \frac{30}{10} = 3 \text{ m}$$

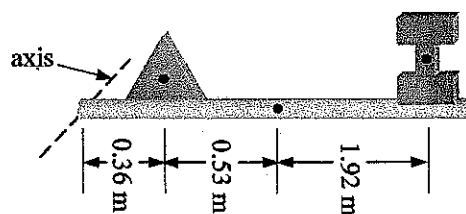
31. Three objects are positioned along the x axis as follows: 4.4 kg at $x = +1.1$ m, 3.7 kg at $x = -0.80$ m, and 2.9 kg at $x = -1.6$ m. The acceleration due to gravity is the same everywhere. What is the distance from the location of the center of gravity to the location of the center of mass for this system?

- (a) zero meters (c) -0.26 m (e) +0.52 m
(b) -0.52 m (d) +0.26 m

Center of gravity = Center of mass
if $g = \text{constant everywhere}$

A

32. A 14-kg beam is hinged at one end. A 6.0-kg triangular object and a 7.5-kg I-shaped object are positioned as shown. Dots indicate the individual centers of gravity of the beam and the two objects. What is the distance from the axis of rotation to the center of gravity for this system?



- (a) 1.3 m (c) 0.96 m
(b) 1.1 m (d) 0.89 m

(e) 0.71 m

$$\bar{x}_{cm} = \frac{6(0.36) + 14(0.99) + 7.5(2.28)}{27.5} = 1.38$$

Newton's Second Law for Rotational Motion about a Fixed Axis

C

33. Consider the following four objects:

- a hoop - MR^2 a solid sphere - $\frac{2}{5}MR^2$
a flat disk - $\frac{1}{2}MR^2$ a hollow sphere - $\frac{2}{3}MR^2$

Each of the objects has mass M and radius R . The axis of rotation passes through the center of each object, and is perpendicular to the plane of the hoop and the plane of the flat disk. Which object requires the largest torque to give it the same angular acceleration?

- (a) the solid sphere (d) the flat disk
(b) the hollow sphere (e) both the solid and the hollow spheres
(c) the hoop - Has the largest inertia I .

D

34. A 50-N · m torque acts on a wheel with a moment of inertia 150 kg · m². If the wheel starts from rest, how long will it take the wheel to make one revolution?

- (a) 0.33 s (c) 2.4 s
(b) 0.66 s (d) 6.1 s

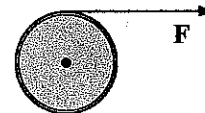
$$\tau = I\alpha \quad 50 = 150\alpha \quad \alpha = \frac{1}{3} \text{ rad/s}^2$$

$$\Delta\theta = \omega_i t + \frac{1}{2}\alpha t^2 \quad t^2 = 12\pi$$

$$2\pi = \frac{1}{2}\alpha t^2 \quad t = 6.14$$

C

35. A string is wrapped around a pulley of radius 0.05 m and moment of inertia 0.2 kg · m². If the string is pulled with a force F , the resulting angular acceleration of the pulley is 2 rad/s². Determine the magnitude of the force F .

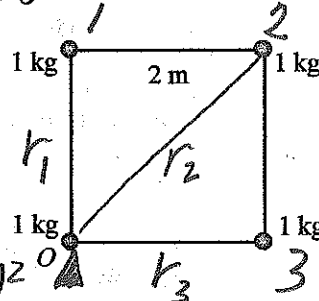


- (a) 0.4 N (c) 8 N (e) 40 N
(b) 2 N (d) 16 N

$$\tau = I\alpha = r \times F \quad 2(0.2) = 0.05(F) \sin 90^\circ \quad F = 8$$

E

36. A massless frame in the shape of a square with 2-m sides has a 1-kg ball at each corner. What is the moment of inertia of the four balls about an axis through the corner marked O and perpendicular to the plane of the paper?



- (a) 4 kg · m²
(b) 8 kg · m²
(c) 10 kg · m²
(d) 12 kg · m²
(e) 16 kg · m²

$$I = \sum mr^2$$

$$I = (1)(2)^2 + (1)(2^2 + 2^2) + (1)(2)^2$$

$$= 4 + 8 + 4 = 16$$

B

37. A certain merry-go-round is accelerated uniformly from rest and attains an angular speed of 0.4 rad/s in the first 10 seconds. If the net applied torque is 2000 N · m, what is the moment of inertia of the merry-go-round?

- (a) 400 kg · m² (d) 800 kg · m²
(b) 50 000 kg · m² (e) This cannot be determined since the radius is not specified.
(c) 5000 kg · m²

$$\tau = I\alpha \quad \omega_f = \omega_i + \alpha t$$

$$2000 = I(0.04) \quad 0.4 = \alpha(10)$$

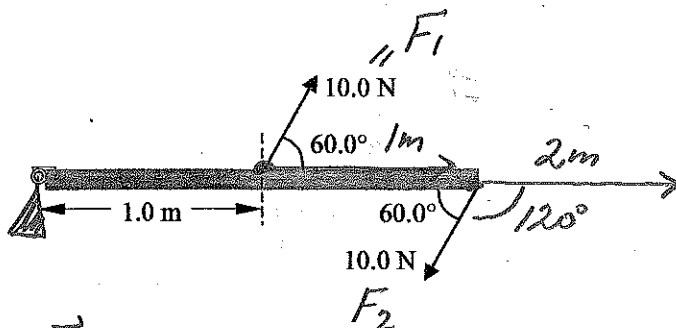
$$I = 50,000 \text{ kg} \cdot \text{m}^2 \quad \alpha = 0.04$$

C

38. The drawing shows the top view of a door that is 2 m wide. Two forces are applied to the door as indicated.

What is the magnitude of the net torque on the door with respect to the hinge?

- (a) 0 N · m (d) 10.0 N · m
(b) 5.0 N · m (e) 26.0 N · m
(c) 8.7 N · m



$$\vec{\tau}_1 = (1)(10\text{ N})\sin(60^\circ)(+\hat{z}) = 8.66$$

$$\vec{\tau}_2 = (2)(10\text{ N})\sin(120^\circ)(-\hat{z}) = -17.32$$

$$\tau_{\text{net}} = -17.32 + 8.66 = -8.66 \text{ N}\cdot\text{m}$$

B

39. Which one of the following statements concerning the moment of inertia I is false?

- (a) I may be expressed in units of $\text{kg} \cdot \text{m}^2$. — Yes $I = mr^2$
(b) I depends on the angular acceleration of the object as it rotates. — $I = \text{sum of } mr^2$
(c) I depends on the location of the rotation axis relative to the particles that make up the object. — Yes
(d) I depends on the orientation of the rotation axis relative to the particles that make up the object. — Yes
(e) Of the particles that make up an object, the particle with the smallest mass may contribute the greatest amount to I . — Yes if r^2 is big enough

A

40. Two uniform solid spheres, A and B have the same mass. The radius of sphere B is twice that of sphere A. The axis of rotation passes through each sphere. Which one of the following statements concerning the moments of inertia of these spheres is true?

- (a) The moment of inertia of A is one-fourth that of B.
(b) The moment of inertia of A is one-half that of B.
(c) The moment of inertia of A is 5/4 that of B.
(d) The moment of inertia of A is 5/8 that of B.
(e) The two spheres have equal moments of inertia.

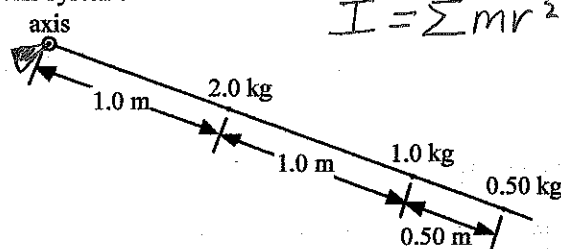
$$I = \frac{2}{5}MR^2 \quad I_B = \frac{2}{5}M(2R_A)^2$$

$$R_B = 2R_A \quad I_B = \frac{2}{5}M(4R_A^2)$$

$$I_B = I_A(4)$$

E

41. Three objects are attached to a massless rigid rod that has an axis of rotation as shown. Assuming all of the mass of each object is located at the point shown for each, calculate the moment of inertia of this system.



$$I = \sum mr^2 = (2)(1)^2 + 1(2)^2 + 0.5(2.5)^2$$

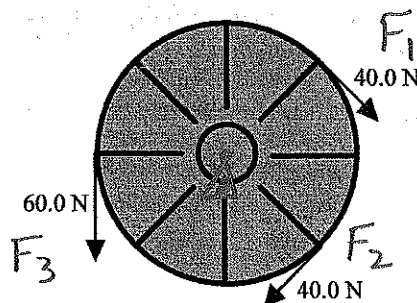
$$= 2 + 4 + 3.125 = 9.125$$

- (a) 1.3 $\text{kg} \cdot \text{m}^2$ (c) 5.3 $\text{kg} \cdot \text{m}^2$ (e) 9.1 $\text{kg} \cdot \text{m}^2$
(b) 3.1 $\text{kg} \cdot \text{m}^2$ (d) 7.2 $\text{kg} \cdot \text{m}^2$

B

42. Three children are pulling on a rotatable platform on a playground. The platform has a radius of 3.65 m. In the picture, two children are pulling with equal forces of 40.0 N in an attempt to make the platform rotate clockwise. The third child applies a force of 60 N as shown. What is the net torque on the platform?

Note: "ccw" is counterclockwise and "cw" is clockwise.



- (a) 73 N · m, ccw (c) 511 N · m, ccw
(b) 73 N · m, cw (d) 511 N · m, cw

$$\tau_{\text{net}} = \tau_1 + \tau_2 + \tau_3$$

$$= -(3.65)(40) - (3.65)(40) + 3.65(60)$$

$$= -146 - 146 + 219 = -73 \hat{z}$$

$$\tau = I\alpha \quad .1(12) = .15(\alpha) \quad \alpha = 8 \text{ rad/s}^2$$

E

43. A string is wrapped around a pulley of radius 0.10 m and moment of inertia 0.15 kg · m². The string is pulled with a force of 12 N. What is the magnitude of the resulting angular acceleration of the pulley?

- (a) 18 rad/s² (c) 80 rad/s² (e) 8.0 rad/s²
 (b) 0.13 rad/s² (d) 0.055 rad/s²

E

44. A 45-N brick is suspended by a light string from a 2.0-kg pulley. The brick is released from rest and falls to the floor below as the pulley rotates through 5.0 rad. The pulley may be considered a solid disk of radius 1.5 m. What is the angular speed of the pulley?

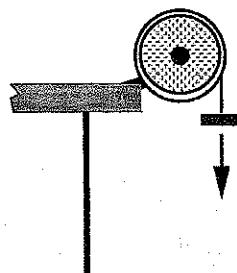
- (a) 17 rad/s (d) 8.1 rad/s
 (b) 15 rad/s (e) 7.3 rad/s
 (c) 9.4 rad/s

$$M_{\text{brick}} = \frac{45}{9.8} = 4.6 \text{ kg}$$

$$\tau = I\alpha$$

$$1.5(T) = 2.25(\alpha)$$

$$T = 1.5\alpha$$



$$I = \frac{1}{2}MR^2 = \frac{1}{2}(2)(1.5)^2 = 2.25$$

$$F = ma$$

$$T - 45 = -4.6[\alpha(1.5)]$$

$$1.5\alpha - 45 = -6.9\alpha$$

Rolling Motion

B

45. A bicycle with wheels of radius 0.4 m travels on a level road at a speed of 8 m/s. What is the angular speed of the wheels?

- (a) 10 rad/s (c) $(\pi/10)$ rad/s (e) $(20/\pi)$ rad/s
 (b) 20 rad/s (d) (10π) rad/s

$$v_{cm} = \omega R \quad \omega = 8/0.4 = 20 \text{ rad/s}$$

$$\alpha = 5.36 \text{ rad/s}^2$$

$$\omega_f^2 = \omega_i^2 + 2\alpha\Delta\theta$$

$$= 2(5.36)5$$

$$\omega = \sqrt{53.6} = 7.3$$

D

46. A bicycle has tires of radius 0.35 meters. If the bicycle is traveling at a constant speed of 12 m/s, at approximately what angular speed are the tires rotating?

- (a) 85 rev/min (c) 214 rev/min (e) 423 rev/min
 (b) 197 rev/min (d) 327 rev/min

$$v_{cm} = \omega R \quad \omega = 12/0.35 = 34 \text{ rad/s} \cdot \frac{1 \text{ rev}}{2\pi \text{ rad}} \cdot \frac{60 \text{ s}}{1 \text{ min}} = 324$$

C

47. A circular hoop rolls without slipping on a flat horizontal surface. Which one of the following is necessarily true?

- (a) All points on the rim of the hoop have the same speed. — No
 (b) All points on the rim of the hoop have the same velocity. — No
 (c) Every point on the rim of the wheel has a different velocity. — Yes $v_{cm} + \omega R$
 (d) All points on the rim of the hoop have acceleration vectors that are tangent to the hoop. — No
 (e) All points on the rim of the hoop have acceleration vectors that point toward the center of the hoop. — No, wheel can be accelerating linearly a_{cm}

D

48. A bicycle wheel of radius 0.70 m is turning at an angular speed of 6.3 rad/s as it rolls on a horizontal surface without slipping. What is the linear speed of the wheel?

- (a) 1.4 m/s (b) 0.11 m/s (e) 9.1 m/s
 (b) 28 m/s (d) 4.4 m/s

$$v_{cm} = \omega R = 6.3 \text{ rad/s} (0.7 \text{ m}) = 4.41 \text{ m/s}$$

Questions 49 and 50 pertain to the following situation:

$$\frac{2 \text{ rev}}{5} = \frac{2\pi \text{ rad}}{1 \text{ rev}} = 4\pi \frac{\text{rad}}{5} = \omega_i$$

A bicycle wheel of radius 0.70 m is rolling without slipping on a horizontal surface with an angular speed of 2.0 rev/s when the cyclist begins to uniformly apply the brakes. The bicycle stops in 5.0 s.

D

49. Through how many revolutions did the wheel turn during the 5.0 seconds of braking?

- (a) 10 rev (c) 9.6 rev (e) 0.4 rev
(b) 2.0 rev (d) 5.0 rev

C

50. How far did the bicycle travel during the 5.0 seconds of braking?

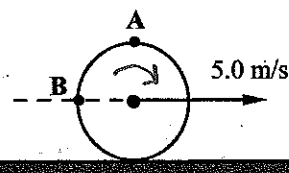
- (a) 1.8 m (c) 22 m (e) 44 m
(b) 8.8 m (d) 42 m

$$\begin{aligned} v_{cm} &= \omega r = 4\pi(0.7) = 8.8 \frac{\text{m}}{\text{s}} \\ v_f &= v_i + at \quad a = \frac{-8.8}{5} = -1.76 \frac{\text{m}}{\text{s}^2} \\ \alpha &= \frac{a}{r} = \frac{-1.76}{0.7} = -2.514 \\ \omega_f^2 &= \omega_i^2 + 2\alpha\Delta\theta \\ 0 &= (4\pi)^2 - 2(2.514)\Delta\theta \\ \Delta\theta &= 31.4 \text{ rad} \\ &\text{or} \\ &5 \text{ rev} \end{aligned}$$

$$x_f = x_i + v_i t + \frac{1}{2}at^2 = 8.8(5) - \frac{1}{2}(1.76)5^2 = 22 \text{ m}$$

Questions 52 through 55 pertain to the statement below:

A 2.0-kg solid disk rolls without slipping on a horizontal surface so that its center proceeds to the right with speed 5.0 m/s.



The point A is the uppermost point on the disk and the point B is along the horizontal line that connects the center of the disk to the rim.

C

51. What is the direction of the disk's angular velocity?

- (a) to the left (d) out of the paper
(b) to the right (e) It varies from point to point on the disk.
(c) into the paper

Curl Right hand fingers stick out thumb

E

52. What is the instantaneous speed of point A with respect to the ground?

- (a) zero m/s (c) 5.0 m/s (e) 10.0 m/s
(b) 2.5 m/s (d) 7.1 m/s

$$v_{top} = 2v_{cm} = 2(5) = 10 \text{ m/s}$$

C

53. What is the instantaneous speed of point B with respect to the ground?

- (a) zero m/s (c) 7.1 m/s (e) 10.0 m/s
(b) 5.0 m/s (d) 7.5 m/s

$$\vec{v} = \vec{v}_x + \vec{v}_{tan} = 5\hat{x} + 5\hat{y} \quad v = \sqrt{5^2 + 5^2} = 7.07 \text{ m/s}$$

A

54. What is the instantaneous speed of the point of the disk that makes contact with the surface?

- (a) zero m/s (c) 7.1 m/s (e) 10.0 m/s
(b) 5.0 m/s (d) 7.5 m/s

$$v_{bottom} = 0 \quad \text{No slipping}$$

B

Rotational Work and Energy

55. A hollow cylinder of mass M and radius R rolls down an inclined plane. A block of mass M slides down an identical inclined plane. Complete the following statement: If both objects are released at the same time,

- (a) the cylinder will reach the bottom first. — No, rolling takes energy
 (b) the block will reach the bottom first. — Yes, all PE converted to translational KE
 (c) the block will reach the bottom with the greater kinetic energy. — No, Total $KE_f = PE_i$
 (d) the cylinder will reach the bottom with the greater kinetic energy. — No
 (e) both the block and the cylinder will reach the bottom at the same time. — No

A

56. A solid sphere and a hollow sphere each of mass M and radius R are released at the same time from the top of an inclined plane. Which one of the following statements is necessarily true?

- (a) The solid sphere will reach the bottom first.
 (b) The hollow sphere will reach the bottom first.
 (c) Both spheres will reach the bottom at the same time.
 (d) The solid sphere will reach the bottom with the greater kinetic energy.
 (e) The hollow sphere will reach the bottom with the greater kinetic energy.

$I_{\text{solid}} = \frac{2}{5}MR^2$ $I_{\text{hollow}} = \frac{2}{3}MR^2$
 $I_{\text{solid}} < I_{\text{hollow}}$
 ∴ Solid wins

C

57. Consider the following three objects, each of the same mass and radius:

(1) a solid sphere
 $I = \frac{2}{5}MR^2$

(2) a solid disk
 $I = \frac{1}{2}MR^2$

(3) a hoop
 $I = MR^2$

All three are released from rest at the top of an inclined plane. The three objects proceed down the incline undergoing rolling motion without slipping. In which order do the objects reach the bottom of the incline?

- (a) 3, 1, 2 (c) 1, 2, 3 (e) All three reach the bottom at the same time.
 (b) 2, 3, 1 (d) 3, 2, 1

Which I smallest? (1) $\frac{1}{2}MR^2$ (2) $\frac{1}{5}MR^2$ (3) $1MR^2$

D

58. A 50-kg rider on a moped of mass 75 kg is traveling with a speed of 20 m/s. Each of the two wheels of the moped has a radius of 0.2 m and a moment of inertia of $0.2 \text{ kg} \cdot \text{m}^2$. What is the total rotational kinetic energy of the wheels?

- (a) 80 J (c) 500 J (e) 4000 J
 (b) 100 J (d) 2000 J

$KE_{\text{rot}} = \frac{1}{2}I\omega^2 = \frac{1}{2}I\left(\frac{v_{\text{cm}}}{r}\right)^2 = 2 \left[\frac{1}{2}(0.2)\left(\frac{20}{0.2}\right)^2 \right] = 2000 \text{ J}$

C

59. A 1.0-kg wheel in the form of a solid disk rolls along a horizontal surface with a speed of 6.0 m/s.

- What is the total kinetic energy of the wheel?
 (a) 9.0 J (c) 27 J (e) 54 J
 (b) 18 J (d) 36 J

$KE = \frac{1}{2}\left(\frac{1}{2}MR^2\right)\left(\frac{v}{R}\right)^2 + \frac{1}{2}Mv^2 = \frac{1}{4}Mv^2 + \frac{1}{2}Mv^2 = \frac{3}{4}Mv^2 = \frac{3}{4}(1)(6)^2 = 27 \text{ J}$

A

60. A 2.0-kg solid cylinder of radius 0.5 m rotates at a rate of 40 rad/s about its cylindrical axis. What power is required to bring the cylinder to rest in 10 s?

- (a) 20 W (c) 160 W
 (b) 40 W (d) 200 W

$P = \frac{W}{t} = \frac{200 \text{ J}}{10 \text{ s}} = 20 \text{ W}$

(e) 400 W $KE = \frac{1}{2}I\omega^2 = \frac{1}{2}\left[\frac{1}{2}(2)(0.5)^2\right](40)^2 = 200 \text{ J}$

B

61. A solid cylinder of radius 0.35 m is released from rest from a height of 1.8 m and rolls down the incline as shown. What is the angular speed of the cylinder when it reaches the horizontal surface?



- (a) 8.2 rad/s (c) 34 rad/s (e) This cannot be determined because the mass is unknown.
 (b) 14 rad/s (d) 67 rad/s

$v_{\text{cm}} = \omega R$
 $KE_f = PE_i$ $I = \frac{1}{2}MR^2 = \frac{1}{2}M(0.35)^2$
 $\frac{1}{2}Mv_{\text{cm}}^2 + \frac{1}{2}I\omega^2 = mgh \rightarrow \frac{1}{2}M(\omega R)^2 + \frac{1}{2}\left[\frac{1}{2}M(0.35)^2\right]\omega^2 = Mgh$
 $\omega = \left[\frac{9.8(1.8)}{0.35^2 + \frac{(0.35)^2}{4}} \right]^{\frac{1}{2}} = 13.86$

E

62. A solid sphere rolls without slipping along a horizontal surface. What percentage of its total kinetic energy is rotational kinetic energy?

- (a) 33 %
(b) 50 %

- (c) 12 %
(d) 75 %

(e) 29 %

$$\frac{KE_{rot}}{KE_{tot}} = \frac{\frac{1}{2} \left[\frac{2}{5} MR^2 \right] \omega^2}{\frac{1}{2} M (\omega R)^2 + \frac{1}{2} \left[\frac{2}{5} MR^2 \right] \omega^2} = \frac{\frac{1}{5}}{\frac{1}{2} + \frac{1}{5}} = \frac{0.2}{0.7} = 0.29$$

C

63. A hollow sphere of radius 0.25 m is rotating at 13 rad/s about an axis that passes through its center. The mass of the sphere is 3.8 kg. Assuming a constant net torque is applied to the sphere, how much work is required to bring the sphere to a stop?

- (a) 1.0 J
(b) 3.8 J

- (c) 13 J
(d) 25 J

(e) 38 J

$$I = \frac{2}{3} (3.8) (0.25)^2 = .158 \text{ kg} \cdot \text{m}^2$$

E

64. A ceiling fan has five blades, each with a mass of 0.34 kg and a length of 0.66 m. The fan is operating in its "low" setting at which the angular speed is 9.4 rad/s. If the blades can be approximated as uniform thin rods that rotate about one end, what is the total rotational kinetic energy of the five blades?

- (a) 35 J
(b) 29 J

- (c) 23 J
(d) 17 J

(e) 11 J

$$I_{rod} = \frac{1}{3} ML^2 = \frac{1}{3} (0.34) (0.66)^2 = .094 \text{ kg} \cdot \text{m}^2$$

D

65. A solid cylinder with a mass m and radius r is mounted so that it can be rotated about an axis that passes through the center of both ends. At what angular speed ω must the cylinder rotate to have the same total kinetic energy that it would have if it were moving horizontally with a speed v without rotation?

- (a) $\omega = \frac{v}{r}$
(b) $\omega = \frac{v^2}{r^2}$

- (c) $\omega = \frac{v}{2r}$
(d) $\omega = \frac{v}{r} \sqrt{2}$

(e) $\omega = \frac{v^2}{2r}$

$$\frac{1}{2} mv^2 = \frac{1}{2} \left[\frac{1}{2} mr^2 \right] \omega^2$$

$$\omega^2 = \frac{2v^2}{r^2}$$

$$\omega = \sqrt{2} \frac{v}{r}$$

Angular Momentum

D

66. A child standing on the edge of a freely spinning merry-go-round moves quickly to the center. Which one of the following statements is necessarily true concerning this event and why?

- (a) The angular speed of the system decreases because the moment of inertia of the system has increased.
(b) The angular speed of the system increases because the moment of inertia of the system has increased.
(c) The angular speed of the system decreases because the moment of inertia of the system has decreased.
(d) The angular speed of the system increases because the moment of inertia of the system has decreased. *yes*
(e) The angular speed of the system remains the same because the net torque on the merry-go-round is zero N · m.

$$I_i = \frac{1}{2} MR^2 + mr_i^2$$

if $r_f < r_i$

I_f is smaller

$$L_i = L_f \quad I_i \omega_i = I_f \omega_f$$

C

67. What happens when a spinning ice skater draws in her outstretched arms?

- (a) Her angular momentum decreases.
(b) Her angular momentum increases.
(c) Her moment of inertia decreases causing her to speed up.
(d) Her moment of inertia decreases causing her to slow down.
(e) The torque that she exerts increases her moment of inertia.

I decreases since $r_f < r_i$

B

68. A spinning star begins to collapse under its own gravitational pull. Which one of the following occurs as the star becomes smaller?

- (a) The star's angular velocity decreases. *-w increases*
 (b) The star's angular momentum remains constant. *yes, $L_i = L_f$ since no τ_{net} external*
 (c) The star's angular momentum increases.
 (d) The star's angular velocity remains constant.
 (e) Both the star's angular momentum and its angular velocity remain constant.

C

69. A spinning skater draws in her outstretched arms thereby reducing her moment of inertia by a factor of 2. Determine the ratio of her final kinetic energy to her initial kinetic energy.

- (a) 0.5 (c) 2 (e) 16 *$I_f \omega_f = I_i \omega_i$*
 (b) 1 *$KE_i = \frac{1}{2} I \omega_i^2$* (d) 4 *$KE_f = \frac{1}{2} [\frac{1}{2} I] (2\omega_i)^2$* *$\omega_f = \frac{I}{\frac{1}{2} I} \omega_i$*
 $\omega_f = 2\omega_i$

E

70. A 1500-kg satellite orbits a planet in a circular orbit of radius 6.2×10^6 m. What is the angular momentum of the satellite in its orbit around the planet if the satellite completes one orbit every 1.5×10^4 s?

- (a) $3.9 \times 10^6 \text{ kg} \cdot \text{m}^2/\text{s}$ (c) $6.2 \times 10^8 \text{ kg} \cdot \text{m}^2/\text{s}$ (e) $2.4 \times 10^{13} \text{ kg} \cdot \text{m}^2/\text{s}$
 (b) $1.4 \times 10^{14} \text{ kg} \cdot \text{m}^2/\text{s}$ (d) $8.1 \times 10^{11} \text{ kg} \cdot \text{m}^2/\text{s}$

$$L = r m v = 6.2 \times 10^6 (1500) (2597) = 2.4 \times 10^{13}$$

D

71. A 60.0-kg skater begins a spin with an angular speed of 6.0 rad/s. By changing the position of her arms, the skater decreases her moment of inertia by 50%. What is the skater's final angular speed?

- (a) 3.0 rad/s (c) 9.0 rad/s (e) 18 rad/s
 (b) 4.5 rad/s (d) 12 rad/s

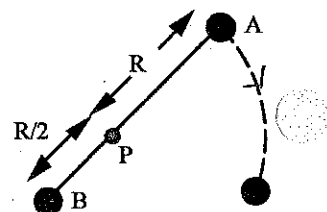
$$I_i \omega_i = I_f \omega_f$$

$$.5 I_i \omega_i = I_i \omega_f$$

$$\omega_f = 12$$

D

72. Two equal spheres, labeled A and B in the figure, are attached to a massless rod with a frictionless pivot at the point P. The system is made to rotate clockwise with angular speed ω on a horizontal, frictionless tabletop. Sphere A collides with and sticks to another equal sphere that is at rest on the tabletop. Note: the masses of all three spheres are equal. What is the angular speed of the system immediately after the collision?



- (a) ω (c) 0.60ω (e) 0.29ω
 (b) 0.82ω (d) 0.56ω

$$(m \frac{R^2}{4} + m R^2) \omega_i = (m \frac{R^2}{4} + m R^2 + m R^2) \omega_f$$

$$\frac{5}{4} \omega_i = \frac{9}{4} \omega_f$$

$$\omega_f = \frac{5}{9} \omega_i = .56$$

E

73. Planets A and B are uniform solid spheres that rotate at a constant speed about axes through their centers. Although B has twice the mass and three times the radius of A, each planet has the same rotational kinetic energy. What is the ratio ω_B / ω_A of their angular speeds?

- (a) 0.055 (c) 0.165 (e) 0.236
 (b) 0.093 (d) 0.191

$$I_{\text{sphere}} = \frac{2}{5} M R^2$$

$$\frac{1}{2} I_A \omega_A^2 = \frac{1}{2} I_B \omega_B^2$$

$$\frac{\omega_B}{\omega_A} = \sqrt{\frac{\frac{2}{5} M_A R_A^2}{\frac{2}{5} M_B R_B^2}} = \sqrt{\frac{1 \cdot 1^2}{2 \cdot 3^2}} = .236$$

E

74. A solid sphere of radius R rotates about a diameter with an angular speed ω . The sphere then collapses under the action of internal forces to a final radius $R/2$. What is the final angular speed of the sphere?

- (a) $\omega/4$ (c) ω (e) 4ω
 (b) $\omega/2$ (d) 2ω

$$I_i = \frac{2}{5} (1) (1)^2 = \frac{2}{5}$$

$$I_f = \frac{2}{5} (1) (\frac{1}{2})^2 = \frac{1}{10}$$

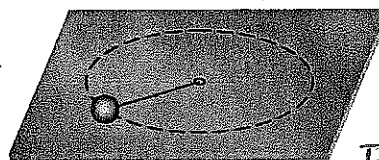
$$I_i \omega = I_f \omega_f$$

$$\omega_f = 4\omega$$

C

75. A ball of mass M moves in a circular path on a horizontal, frictionless surface. It is attached to a light string that passes through a hole in the center of the table. If the string is pulled down, thereby reducing the radius of the path of the ball, the speed of the ball is observed to increase. Complete the following sentence: This occurs because

- (a) the linear momentum of the ball is conserved.
- (b) it is required by Newton's first law of motion.
- (c) the angular momentum of the ball is conserved.
- (d) the angular momentum of the ball must increase.
- (e) the total mechanical energy of the ball must remain constant.



If r decreases
 ω increases

$$L_i = L_f \quad I_i \omega_i = I_f \omega_f$$

$$mR^2 \omega = mr^2 \omega_f$$

$r < R$

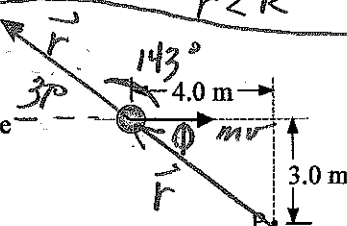
B

76. A 3.0-kg ball moves in a straight line at 10 m/s as shown in the figure. At the instant shown, what is its angular momentum about the point P?

- (a) $30 \text{ kg} \cdot \text{m}^2/\text{s}$
- (b) $90 \text{ kg} \cdot \text{m}^2/\text{s}$
- (c) $120 \text{ kg} \cdot \text{m}^2/\text{s}$
- (d) $150 \text{ kg} \cdot \text{m}^2/\text{s}$
- (e) zero $\text{kg} \cdot \text{m}^2/\text{s}$

$$L = \vec{r} \otimes m\vec{v} = 5(3)(10) \sin 143^\circ$$

$$= 90.27$$

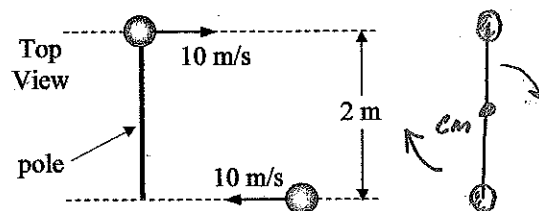


$$\phi = \tan^{-1} \frac{3}{4} = 37^\circ$$

Questions 77 and 78 pertain to the situation described below:

Two skaters, each of mass 40 kg, approach each other along parallel paths that are separated by a distance of 2 m. Both skaters have a speed of 10 m/s.

The first skater carries a 2-m pole that may be considered massless. As he passes the pole, the second skater catches hold of the end. The two skaters then go around in a circle about the center of the pole.



C

77. What is the angular speed of the skaters after they have linked together?

- (a) 4 rad/s
- (b) 5 rad/s
- (c) 10 rad/s
- (d) 20 rad/s
- (e) 40 rad/s

$$L_i = L_f$$

$$2mr v = 2mr(\omega r)$$

$$10 = \omega(1)$$

E

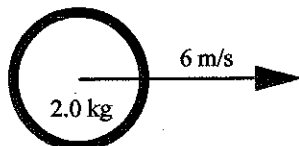
78. What is their combined angular momentum about the center of the pole?

- (a) $2 \text{ kg} \cdot \text{m}^2/\text{s}$
- (b) $40 \text{ kg} \cdot \text{m}^2/\text{s}$
- (c) $80 \text{ kg} \cdot \text{m}^2/\text{s}$
- (d) $400 \text{ kg} \cdot \text{m}^2/\text{s}$
- (e) $800 \text{ kg} \cdot \text{m}^2/\text{s}$

$$L = mrv + mrv = 2(40)(1)(10) = 800$$

Questions 79 and 80 pertain to the situation described below:

A 2.0-kg hoop rolls without slipping on a horizontal surface so that its center proceeds to the right with a constant linear speed of 6.0 m/s.



A

79. Which one of the following statements is true concerning the angular momentum of this hoop?

- (a) It points into the paper.
- (b) It points out of the paper.
- (c) It points to the left.
- (d) It points to the right.
- (e) It varies from point to point on the hoop.

Same direction as ω

$$KE_{total} = \frac{1}{2} I \omega^2 + \frac{1}{2} M v_{cm}^2 = \frac{1}{2} (MR^2) \frac{v_{cm}^2}{R^2} + \frac{1}{2} M v_{cm}^2 = M v_{cm}^2$$

C

80. What is the total kinetic energy of the hoop?

- (a) 36 J
(b) 54 J

- (c) 72 J
(d) 96 J

(e) $140 J = 2(6)^2 = 72$

Additional Problems

D

81. A compact disc rotates about its center at constant angular speed. Which one of the following quantities is constant and non-zero for a dust particle near the edge of the disc?

- (a) linear velocity - *no cm movement*
(b) torque about the center of the disc
(c) centripetal acceleration - $\frac{v^2}{r} = \omega^2 r = \text{const magnitude}$

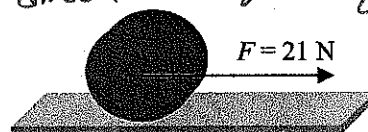
(d) angular momentum - $I\omega = MR^2\omega = \text{constant}$
(e) angular acceleration - $\alpha = 0$
mag & dir
but direction changes in Cartesian coord

B

82. A steady horizontal force F of magnitude 21 N is applied at the axle of a solid disk as shown. The disk has mass 2.0 kg and diameter 0.10 m. What is the linear speed of the center of the disk after it has moved 12 m?

- (a) 9.0 m/s
(b) 13 m/s

- (c) 16 m/s
(d) 22 m/s



- (e) 44 m/s

$W = F \cdot \Delta x = 21 N (12 m)$

$252 = \frac{1}{2} I \omega^2 + \frac{1}{2} M v_{cm}^2$
 $= \frac{1}{2} \left(\frac{1}{2} M R^2 \right) \frac{v_{cm}^2}{R^2} + \frac{1}{2} M v_{cm}^2$

$W = 252 J$

$\uparrow$ Shared w/ rolling

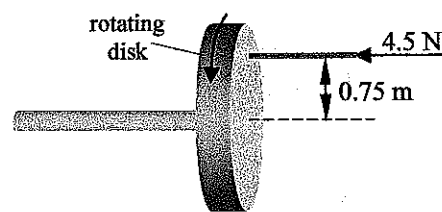
$252 = .75 M v_{cm}^2 \rightarrow v_{cm} = 12.96$

C

83. A uniform disk of radius 1.2 m and mass 0.60 kg is rotating at 25 rad/s around an axis that passes through its center and is perpendicular to the disk. A rod makes contact with the rotating disk with a force of 4.5 N at a point 0.75 m from the axis of rotation as shown. The disk is brought to a stop in 5.0 s. What is the coefficient of kinetic friction for the two materials in contact?

- (a) 0.22
(b) 0.15

- (c) 0.64
(d) 0.37



- (e) 0.55

$\tau = \vec{r} \otimes \vec{F} = .75 (\mu 4.5) = I \alpha$

$\mu (3.375) = (.432)(5)$

$\mu = .64$

$I = \frac{1}{2} M R^2$
 $= \frac{1}{2} (.6)(1.2)^2$

$I = .432$

$\omega_f = \omega_i + \alpha t$

$0 = 25 - \alpha (5)$

$\alpha = 5$

ANSWER KEY

1. B		41. E		81. D
2. C		42. B		82. B
3. C		43. E		83. C
4. A		44. <i>AE</i>		
5. D		45. B		
6. A		46. D		
7. D		47. C		
8. A		48. D		
9. D		49. D		
10. E		50. C		
11. D		51. C		
12. C		52. E		
13. D		53. C		
14. B		54. A		
15. A		55. B		
16. D		56. A		
17. E		57. C		
18. B		58. D		
19. A		59. C		
20. C		60. A		
21. D		61. B		
22. C		62. E		
23. D		63. C		
24. B		64. E		
25. D		65. D		
26. A		66. D		
27. B		67. C		
28. E		68. B		
29. E		69. C		
30. C		70. E		
31. A		71. D		
32. A		72. D		
33. C		73. E		
34. D		74. E		
35. C		75. C		
36. E		76. B		
37. B		77. C		
38. C		78. E		
39. B		79. A		
40. A		80. C		

